

Graphs and Planner Graphs

Outline

- Introduction
- Basic terminology
- Multigraphs and Weighted Graphs
- Representation of Graphs
- Paths and Circuits
- Graph Traversal
- Eulerian Paths and Circuits
- Hamiltonian Paths and Circuits

Motivation for learning graph

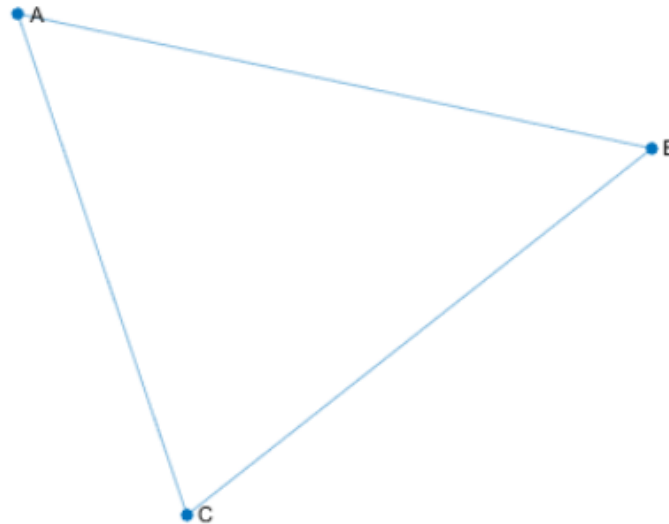
- Graph : Visual representation of relation.
- Many real life problems can be effectively represented via the concept of relations.
- When the number of objects participating in the relation are more, it is convenient to visualize it in the graphical format.
- Ex : Consider a number of cities connected by highways, given a map of highways, we might want to determine whether there is a highway route between two cities of the map.

Introduction

- A graph is a collection of *nodes* and *edges* that represents relationships:
 - **Nodes** are vertices that correspond to objects.
 - **Edges** are the connections between objects.
- For instance, you can model the friendships in a social network using a graph. The graph nodes are people, and the edges represent friendships.

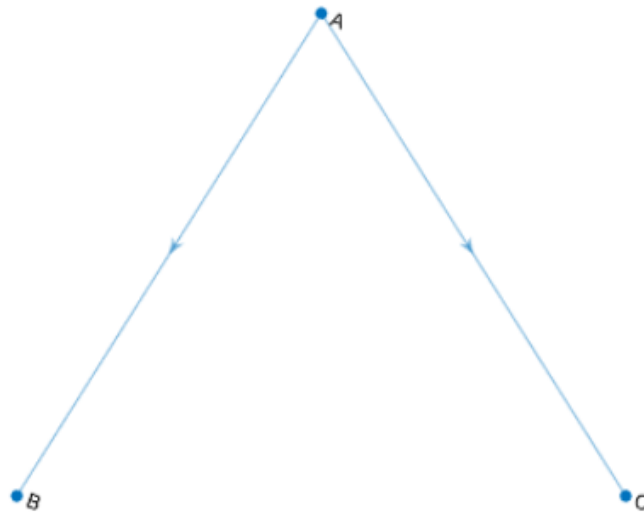
Undirected graphs

- **Undirected graphs** have edges that do not have a direction. The edges indicate a *two-way* relationship, in that each edge can be traversed in both directions.



Directed graphs

- **Directed graphs** have edges with direction. The edges indicate a *one-way* relationship, in that each edge can only be traversed in a single direction.

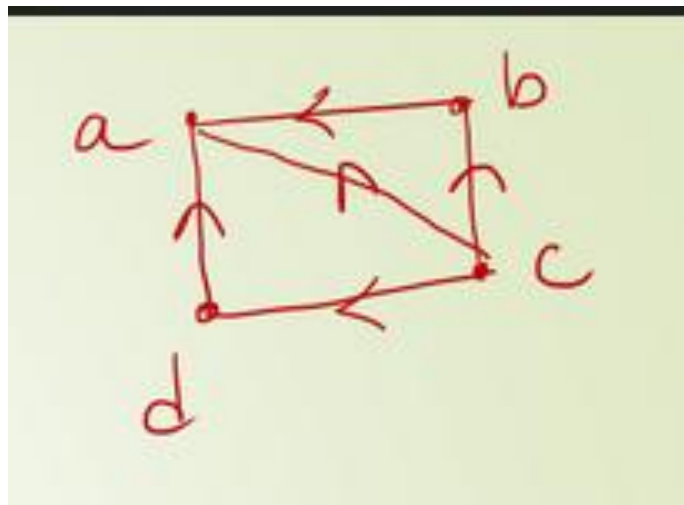


Directed graphs

- Directed graph is defined as pair (V,E) where V is set of vertices .and E is a binary relation on V .
- In undirected graph, E is a set of multisets of the elements from V .

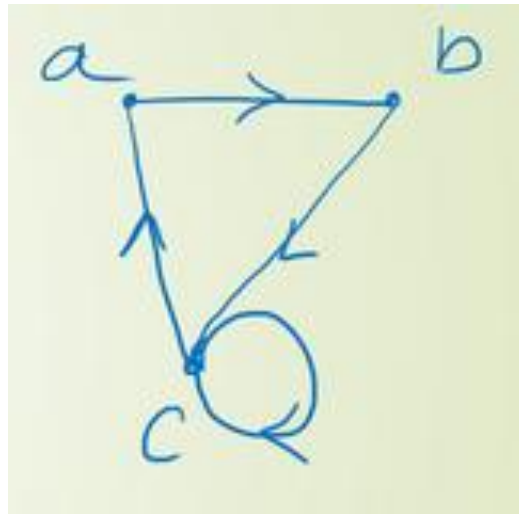
Simple Graph

- Simple Graph : A simple graph is a **graph that does not have more than one edge between any two vertices and no edge starts and ends at the same vertex**. In other words a simple graph is a graph without loops and multiple edges.



Asymmetric digraph

- **Digraphs that have at most one directed edge between a pair of vertices, but are allowed to have self-loops are called asymmetric or anti-symmetric.**



Symmetric digraph

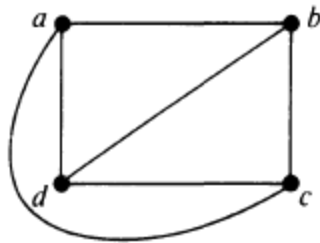
- Digraphs in which for every edge (a,b) , there is also an edge (b,a) .



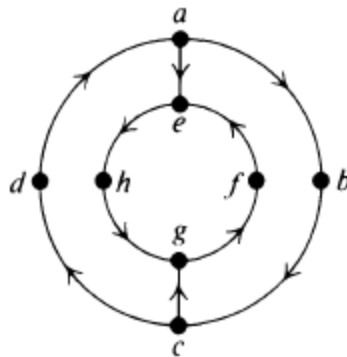
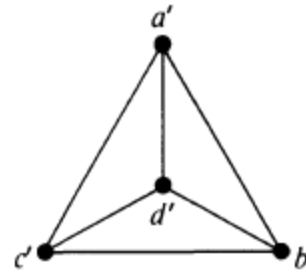
Isomorphic

- A graph can exist in different forms having the same number of vertices, edges, and also the same edge connectivity. Such graphs are called isomorphic graphs.
- Two graphs G_1 and G_2 are said to be isomorphic if –
 - Their number of components (vertices and edges) are same.
 - Their edge connectivity is retained.

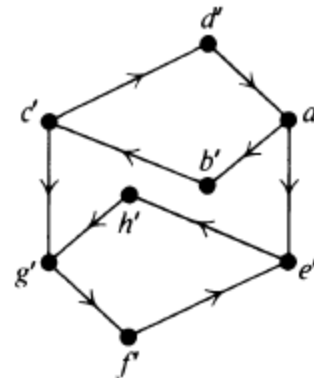
Isomorphic - Example



(a)



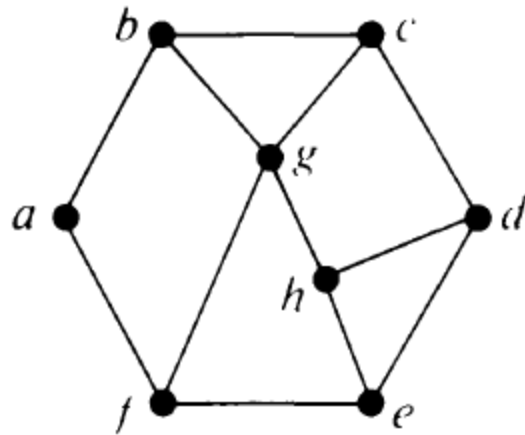
(b)



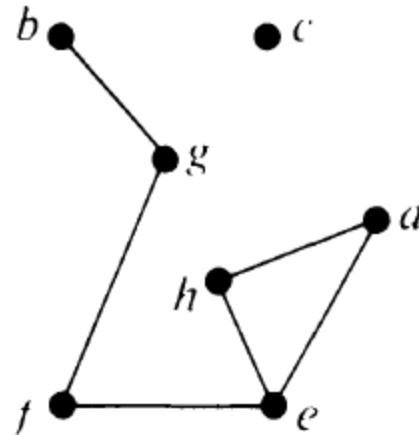
Subgraph

- **Definition:** A *graph* whose *vertices* and *edges* are *subsets* of another *graph*.
- **Formal Definition:** A *graph* $G'=(V', E')$ is a subgraph of another graph $G=(V, E)$ iff
 - $V' \subseteq V$
 - $E' \subseteq E$

Subgraph Example



(a)

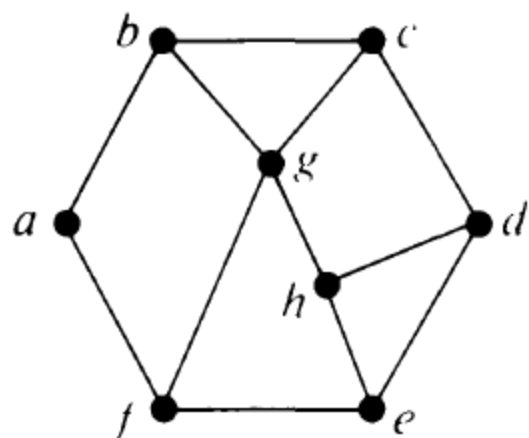


(b)

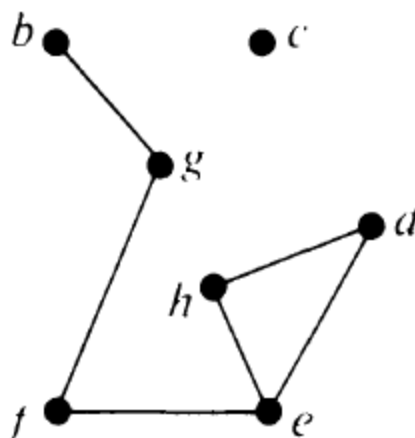
Spanning subgraph

- A subgraph of G is said to be a spanning subgraph if it contains all the vertices of G .

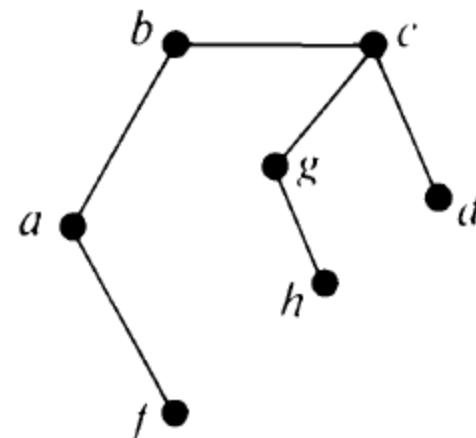
Complement of subgraph



(a)



(b)



(c)

The *complement* of a subgraph $G' = (V', E')$ with respect to the graph G is another subgraph $G'' = (V'', E'')$ such that E'' is equal to $E - E'$ and V'' contains only the vertices with which the edges in E'' are incident.

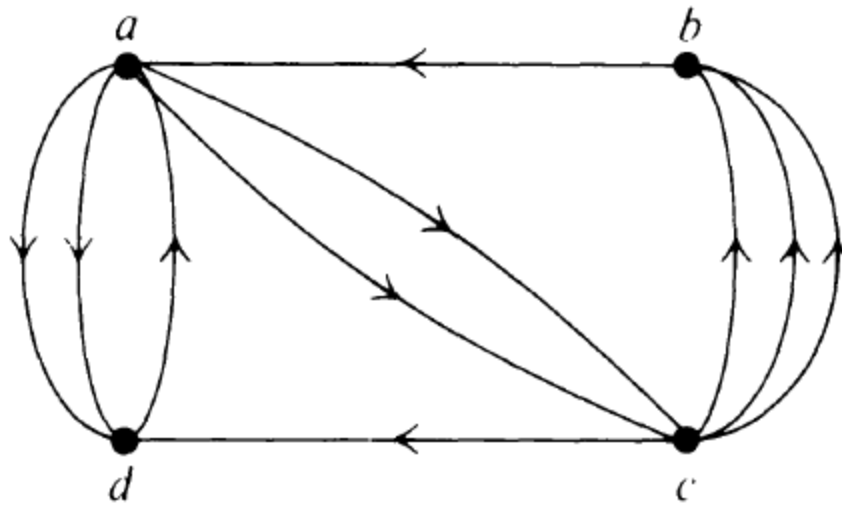
Complete Graph

- Undirected complete graph : Graph of n vertices denoted by K_n is a graph with n vertices in which there is an edge between each pair of vertices.
- Directed Complete Graph : It is a graph with n vertices in which there is exactly one arrow between each pair of distinct vertices.

Multigraph

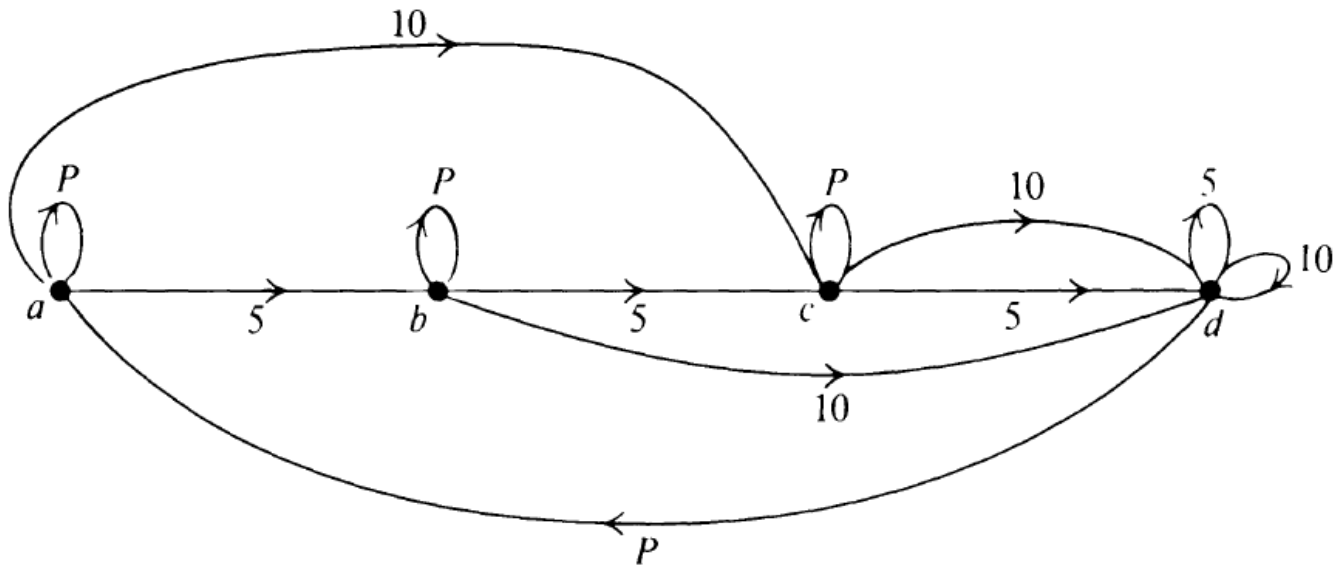
- A multigraph is a **graph which is permitted to have multiple edges** (also called parallel edges), that is, edges that have the same end nodes. Thus two vertices may be connected by more than one edge.
- Let $G(V, E)$, where V is a set and E is a multiset of ordered pairs from $V \times V$. G is called a directed multigraph.

Multigraph



Weighted Graph

- A **graph** whose **edges** have **weights**

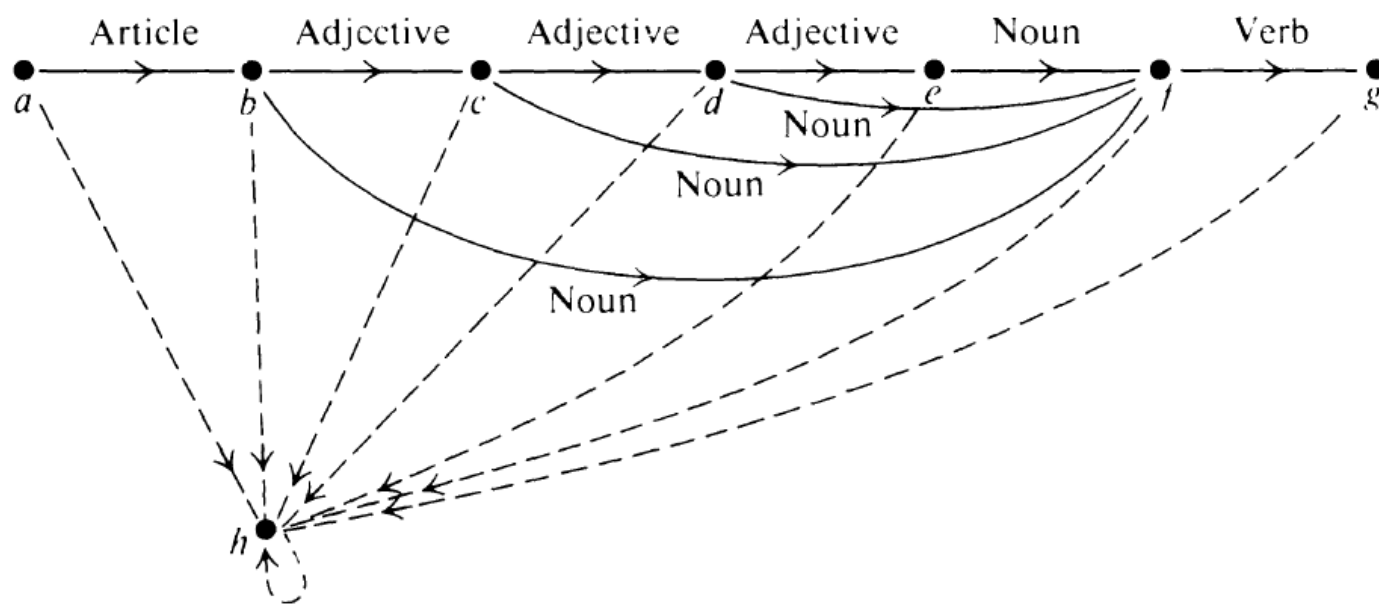


Example 5.2 We consider the problem of recognizing sentences consisting of an article, followed by at most three adjectives, followed by a noun, and then followed by a verb, as shown in the following.

The train stops.

A little girl laughs.

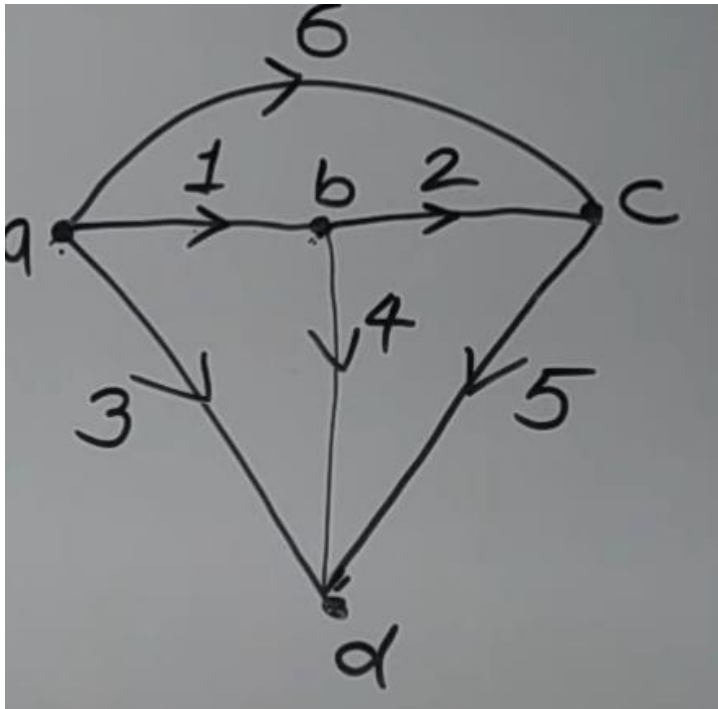
The large fluffy white clouds appear.



Representation of Graph (Incidence Matrix)

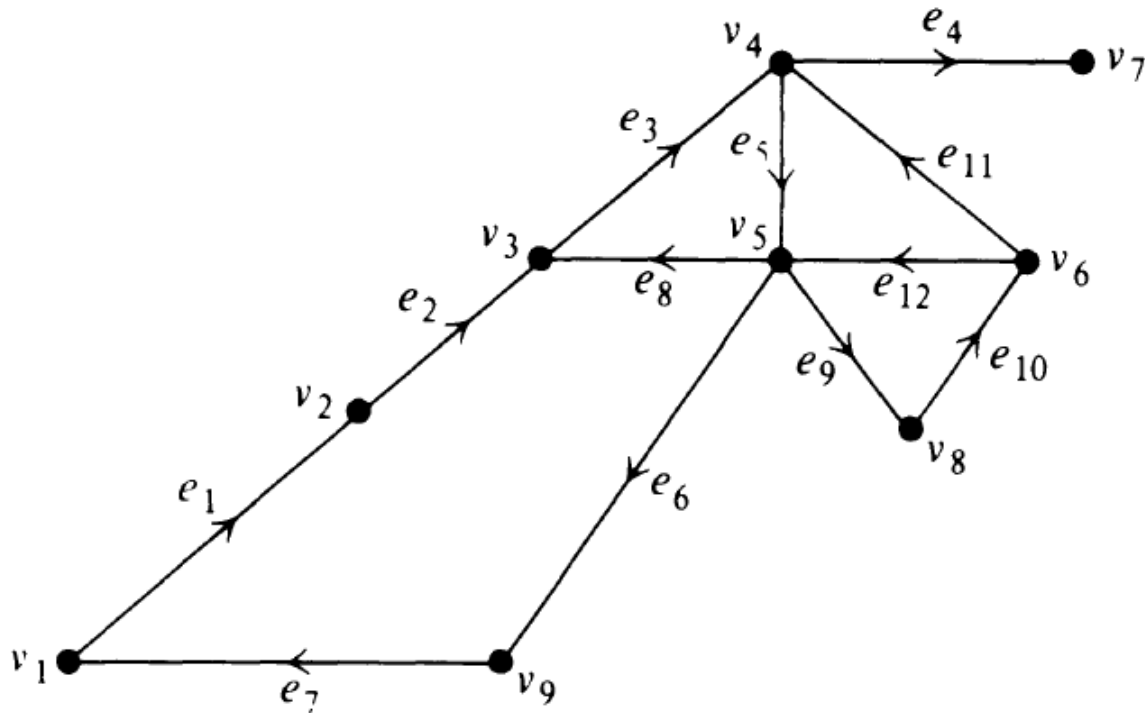
Rows = Vertices

Columns = Edges

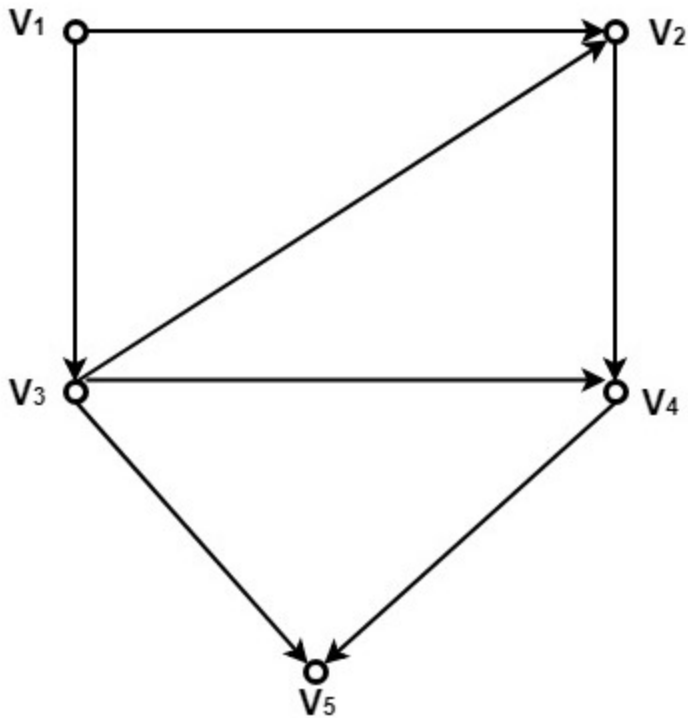


	1	2	3	4	5	6
a	+1	0	+1	0	0	+1
b	-1	+1	0	+1	0	0
c	0	-1	0	0	+1	-1
d	0	0	-1	-1	-1	0

Do it yourself : Represent below graph using incidence matrix and adjacency matrix



Representation of Graph (Adjecancy Matrix)



	V ₁	V ₂	V ₃	V ₄	V ₅
V ₁	0	1	1	0	0
V ₂	0	0	0	1	0
V ₃	0	1	0	1	1
V ₄	0	0	0	0	1
V ₅	0	0	0	0	0